

OCTAVE RING GEOMETRY

A Derived Mapping from the Linear Octave Axis to a Closed Circular Spectrum

Seraphim Framework · v18.3 · Paper 2 v16 · Degeneracy v3

Fundamental Constants

All results in this document follow from four confirmed constants and one topological invariant.

Symbol	Value	Units	Description
n_flip	3.561	octaves	Realm boundary — the linear coupling anchor
n_BBH	5.314	octaves	BBH octave depth (264 events, 0 free params, 0.05σ)
Δn_Ser	1.753	octaves	Seraphim interval = n_BBH – n_flip
N_Hub	201.87	octaves	Planck → Hubble horizon span
χ(S ²)	2	integer	Euler characteristic of the 2-sphere (Euler, 1758)

1. Motivation

The Seraphim framework uses the octave coordinate

$$n = \log_2(v_P / v)$$

as a universal ruler spanning the full physical spectrum from the Planck frequency to the Hubble horizon. This coordinate is intrinsically linear, but many structural relations—antipodes, dualities, and scale correspondences—become clearer when the same axis is represented as a closed ring.

The Octave Ring Geometry formalizes this mapping. No new parameters are introduced; all identities follow from the constants tabulated above.

2. The Linear Octave Axis

Define the global octave interval:

$$n \in [0, N_{\text{Hub}}] \quad \text{where} \quad N_{\text{Hub}} = 201.87$$

Every physical scale corresponds to a unique point on this axis. The axis is equipped with the standard metric

$$d(n_1, n_2) = |n_1 - n_2|$$

3. Curling the Axis into a Ring

Define the equivalence relation

$$n_1 \sim n_2 \iff n_1 - n_2 \in N_{\text{Hub}} \times \mathbb{Z}$$

The quotient space

$$\sim = [0, N_{\text{Hub}}] / \square$$

is a 1-dimensional compact manifold diffeomorphic to S^1 . Map the linear coordinate n to an angular coordinate θ :

$$\theta(n) = 2\pi \times (n / N_{\text{Hub}}) \tag{3.1}$$

Embed $\mathcal{R}$ into the plane via

$$x(n) = R \cos\theta(n), \quad y(n) = R \sin\theta(n) \quad \text{with} \quad R = \frac{N_{\text{Hub}}}{2\pi} \tag{3.2}$$

This choice of R ensures that arc length equals octave distance: $\text{arc}(n_1, n_2) = |n_1 - n_2|$. The transformation preserves all scale distances while revealing global symmetries.

4. The Antipodal Map

Two points separated by half the cavity length lie opposite each other on the ring. The antipode of any octave coordinate n is:

$$A(n) = n + N_{\text{Hub}}/2 \pmod{N_{\text{Hub}}} \quad \text{where} \quad N_{\text{Hub}}/2 = 100.935 \tag{4.1}$$

In angular form, $A(n)$ shifts the phase by π :

$$\theta(A(n)) = \theta(n) + \pi$$

This construction is the mathematical expression of “two sides of the same flat space” when viewed on the ring.

5. The Euler Characteristic Ladder $\chi(S^2)$

The integer spacing of the χ -ladder is fixed by Euler's topological invariant for the 2-sphere. Euler's polyhedral formula (1758) states that for any convex polyhedron:

$$V - E + F = 2$$

In the continuous limit this becomes the Euler characteristic of S^2 :

$$\chi(S^2) = 2$$

This is a topological invariant—it does not depend on the geometry, only on the topology. Within the Seraphim framework, the spin-network area spectrum of Loop Quantum Gravity selects the 2-sphere as the natural patch topology, and $\chi(S^2) = 2$ sets the ladder spacing.

The χ -ladder is defined as:

$$n_k = n_{\text{flip}} + \chi(S^2) \times k = 3.561 + 2k \quad k = 0, 1, 2, \dots \quad (5.1)$$

Every rung n_k corresponds to a geometrically selected scale. The main Seraphim coupling formula uses a related but distinct spacing (3.506 per coupling unit C) derived from the gravitational-wave dataset; the integer ladder (5.1) is the pure topological skeleton underlying it.

Appendix X — Formal Derivation

Lemma $\rightarrow$ Theorem $\rightarrow$ Corollary structure. All results follow from the constants in Section 0 and Lemma 1.

Lemma 1 — Octave Ring and Antipode Map

Let $n \in [0, N_{\text{Hub}}]$ be the octave coordinate. Define the equivalence relation $n_1 \sim n_2 \Leftrightarrow n_1 - n_2 \in N_{\text{Hub}}\mathbb{Z}$. Then the quotient space $\mathcal{R} = [0, N_{\text{Hub}}]/\sim$ is diffeomorphic to S^1 , with angular coordinate $\theta(n) = 2\pi n/N_{\text{Hub}}$.

Define the antipode operator $A(n) = n + N_{\text{Hub}}/2 \pmod{N_{\text{Hub}}}$. Then A is an involution:

$$A(A(n)) = n$$

Proof. $A(A(n)) = n + N_{\text{Hub}}/2 + N_{\text{Hub}}/2 = n + N_{\text{Hub}} \equiv n \pmod{N_{\text{Hub}}}$. ■

Theorem 1 — CMB–BBH Antipode Identity

Let $n_{\text{BBH}} = 5.314$ and $n_{\text{CMB}} = 108.0$, and let $\Delta n_{\text{Ser}} = n_{\text{BBH}} - n_{\text{flip}} = 1.753$. Then:

$$n_{\text{CMB}} = A(n_{\text{BBH}}) + \Delta n_{\text{Ser}} + \varepsilon \quad |\varepsilon| < 0.002$$

Proof.

$$A(n_{\text{BBH}}) = 5.314 + 100.935 = 106.249$$

$$n_{\text{CMB}} - A(n_{\text{BBH}}) = 108.0 - 106.249 = 1.751$$

$$\Delta n_{\text{Ser}} = 1.753$$

$$\varepsilon = 1.751 - 1.753 = -0.002 \quad \therefore |\varepsilon| < 0.002 \quad \blacksquare$$

This identity is not imposed. It emerges from the ring mapping applied to two independently anchored scales.

Corollary 1 — Desert-Center $\rightarrow$ Axion-Realm Antipode

Let the GUT desert walls be fixed by the $\chi(S^2)$ -ladder:

$$n_{\text{string}} = n_{\text{flip}} + 3\chi = 3.561 + 6 = 9.561$$

$$n_{\text{seesaw}} = n_{\text{flip}} + 8\chi = 3.561 + 16 = 19.561$$

Define the desert center:

$$n_{\text{center}} = (n_{\text{string}} + n_{\text{seesaw}}) / 2 = 14.561$$

Then the antipode of the desert center satisfies:

$$A(n_{\text{center}}) = 14.561 + 100.935 = 115.496$$

The nearest χ -ladder rung is $n_{56} = n_{\text{flip}} + 2 \times 56 = 115.561$, with:

$$|\delta| = |115.496 - 115.561| = 0.065 \text{ octaves}$$

Converting n_{56} to energy via $E = h\nu_P \times 2^{(-n_{56})}$:

$$\nu_P = 1.855 \times 10^3 \text{ Hz (Planck frequency)}$$

$$E_{56} = h\nu_P \times 2^{(-115.561)} = 1.25 \text{ } \mu\text{eV}$$

This lies within the QCD axion search band (approximately 1–40 μeV), targeted by ADMX, HAYSTAC, and ABRACADABRA.¹ $\blacksquare$

¹ ADMX Collaboration, Phys. Rev. Lett. 120, 151301 (2018); HAYSTAC Collaboration, Phys. Rev. Lett. 121, 261302 (2018).

Corollary 2 — The α -QCD Inscribed Rectangle

The antipode of the LIGO/ α scale and the QCD Confinement scale, together with their respective antipodal partners, form a geometrically exact inscribed rectangle on the ring.

The four vertices are:

Physical Scale	n (octaves)	Degrees	Pair
Antipode of α	36.155	64.476°	Pair 1
QCD Confinement	68.800	122.693°	Pair 2
α / LIGO / $1/\alpha$	137.090	244.476°	Pair 1
QCD Antipode	169.735	302.693°	Pair 2

The alternating arc lengths are:

Short arc (Antipode(α) $\rightarrow$ QCD): 32.645 octaves = 58.217°

Long arc (QCD $\rightarrow$ α): 68.290 octaves = 121.783°

Verification of exact antipodal pairing:

$$|\alpha - \text{Antipode}(\alpha)| = |137.090 - 36.155| = 100.935 = N_{\text{Hub}}/2 \quad \checkmark$$

$$|\text{QCD} - \text{QCD Antipode}| = |169.735 - 68.800| = 100.935 = N_{\text{Hub}}/2 \quad \checkmark$$

The arc lengths satisfy the rectangle alternation condition to six decimal places:

$$\text{Short}_1 = \text{Short}_2 = 32.6450 \quad \text{Long}_1 = \text{Long}_2 = 68.2900$$

Pythagorean consequence.

By Thales' theorem, any inscribed angle that subtends a diameter of a circle is a right angle. Since both diagonals of the rectangle are diameters of the ring (length = $N_{\text{Hub}}/2$), every corner of the rectangle subtends a diameter and is therefore a right angle. The Pythagorean theorem follows immediately: for chord sides a , b and diagonal $c = 2R$:

$$a^2 + b^2 = 4R^2 \sin^2(\theta/2) + 4R^2 \cos^2(\theta/2) = 4R^2 = c^2$$

The two diameters (the α -antipode axis and the QCD-antipode axis) cross at an angle of 58.22° at ring center. The physical interpretation of the unoccupied antipode(α) slot at $n = 36.155$ ($E \approx 1.59 \times 10^8$ GeV) remains an open prediction. ■

6. Interpretation

The ring geometry provides:

- A mathematically rigorous compactification of the octave axis onto S^1 .
- A natural definition of antipodal dualities via the involution $A(n)$ of Lemma 1.
- A unified geometric language for comparing scales separated by hundreds of octaves.
- Structural explanations for the CMB–BBH identity (Theorem 1), the desert-center–axion correspondence (Corollary 1), and the α –QCD rectangle (Corollary 2).

No new parameters are introduced. All results follow from n_{flip} , n_{BBH} , $\chi(S^2) = 2$, and N_{Hub} .